

Lab Class NLP:IX

Exercise 1 : Word Mover Distance

The lecture introduced the Word Mover's Distance (WMD) for measuring word vector similarity. WMD finds the minimum cumulative transportation cost to move all words from one sentence to words in another sentence in an embedding space [NLP:III-169]:

$$\text{WMD}(x, y) = \min_{\mathbf{T} \geq 0} \sum_{i,j=1}^n \mathbf{T}_{t_i, t_j} c(\mathbf{t}_i, \mathbf{t}_j)$$

where n is the number of words in the sentences and $\mathbf{T}$ is the transportation matrix and $c(\mathbf{t}_i, \mathbf{t}_j)$ is the cost between word embeddings $\mathbf{t}_i$ and $\mathbf{t}_j$. The cost is defined as the Euclidean distance between the word embeddings [NLP:III-161]:

$$c(\mathbf{t}_i, \mathbf{t}_j) = \|\mathbf{t}_i - \mathbf{t}_j\|_2 = \sqrt{\sum_{d=1}^p (t_{i,d} - t_{j,d})^2}$$

You are given the sentences A, B, and C and the 3-dimensional word vectors $[d_1, d_2, d_3]$ for all occurring words:

- Sentence A: "The **cat** **climbed** the tree."
- Sentence B: "The **feline** **scaled** the tree."
- Sentence C: "The **kitten** **ascended** the tree."

	the	cat	climbed	tree	feline	scaled	kitten	ascended
d_1	0.1	0.4	0.7	1.0	0.35	0.75	0.45	0.77
d_2	0.2	0.5	0.8	1.1	0.55	0.85	0.57	0.87
d_3	0.3	0.6	0.9	1.2	0.65	0.95	0.67	0.97

- (a) Calculate the WMD between Sentence A and Sentence B.

(b) Calculate the WMD between Sentence A and Sentence C.

(c) Which sentence B or C is more similar to Sentence A?

Exercise 2 : Sentence Embeddings

The lecture introduced sentence embeddings as a way to represent sentences in a continuous vector space. One approach to generating sentence embeddings is to average the word embeddings $\mathbf{w}$ of all words in a sentence s :

$$\mathbf{s}_{emb} = \frac{1}{|s|} \sum_{\mathbf{w}_i \in s} \mathbf{w}_i$$

You are given the sentences A, B, and C and the 3-dimensional word vectors $[d_1, d_2, d_3]$ for all occurring words:

- Sentence A: "The **cat** **climbed** the tree."
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	the	cat	climbed	tree	feline	scaled	kitten	ascended
d_1	0.1	0.4	0.7	1.0	0.35	0.75	0.45	0.77
d_2	0.2	0.5	0.8	1.1	0.55	0.85	0.57	0.87
d_3	0.3	0.6	0.9	1.2	0.65	0.95	0.67	0.97

- (a) Calculate the sentence embeddings $\mathbf{s}_{emb}(A)$, $\mathbf{s}_{emb}(B)$, and $\mathbf{s}_{emb}(C)$ for sentence A, B, and C, respectively, using vector averaging.

- (b) Which sentence embedding pair is more similar (**A, B**) or (**A, C**)? For your answer, calculate the Manhattan distance D between the embeddings of the sentences in each pair.

The Manhattan distance between two vectors $\mathbf{x}$ and $\mathbf{y}$ is defined as [NLP:III-161]:

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^p |x_i - y_i|,$$

where p is the number of dimensions in the vectors.

- (c) What are the limitations of the vector averaging approach for generating sentence embeddings?
Name at least two.

- (d) Another approach to measuring the similarity between two embedding representations is to calculate the cosine similarity between them [NLP:III-163]:

$$\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = \frac{\mathbf{s}_1 \cdot \mathbf{s}_2}{\|\mathbf{s}_1\| \cdot \|\mathbf{s}_2\|} = \frac{\sum_{i=1}^m s_{1,i} \cdot s_{2,i}}{\sqrt{\sum_{i=1}^m s_{1,i}^2} \cdot \sqrt{\sum_{i=1}^m s_{2,i}^2}}$$

- (d1) Interpret the following cosine similarity values between two sentence embeddings:

a) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = -1$

b) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = 0$

c) $\text{sim}_{\text{cosine}}(\mathbf{s}_1, \mathbf{s}_2) = 1$

- (d2) Calculate the cosine similarity between sentence A and sentence B, and between sentence A and sentence C. Which sentence is more similar to sentence A according to the cosine similarity measure?